

Sailfort Motors

Employee Retention Study

ISSUE / PROBLEM

Sailfort Motors is looking to improve employee retention and investigate **what factors might be causing employees to leave the company.**

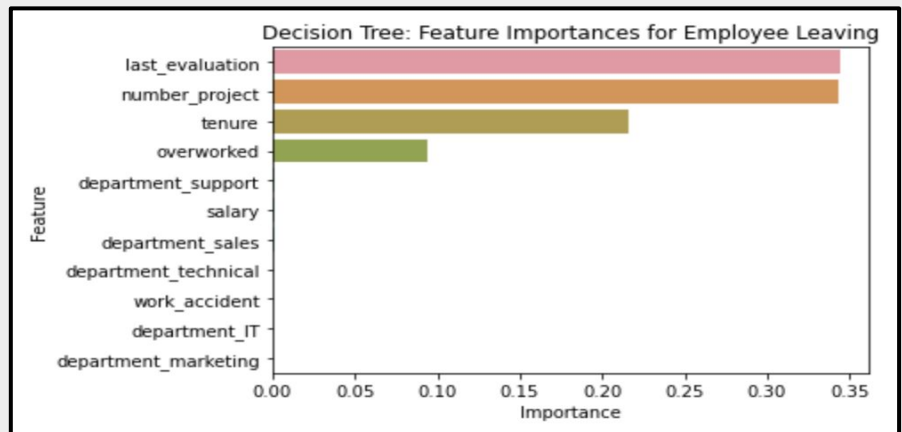
RESPONSE

Due to the target variable I was predicting is categorical (binary), it was necessary to build either a Logistics Regression model or a Tree-Based model (Decision Tree, Random Forest).

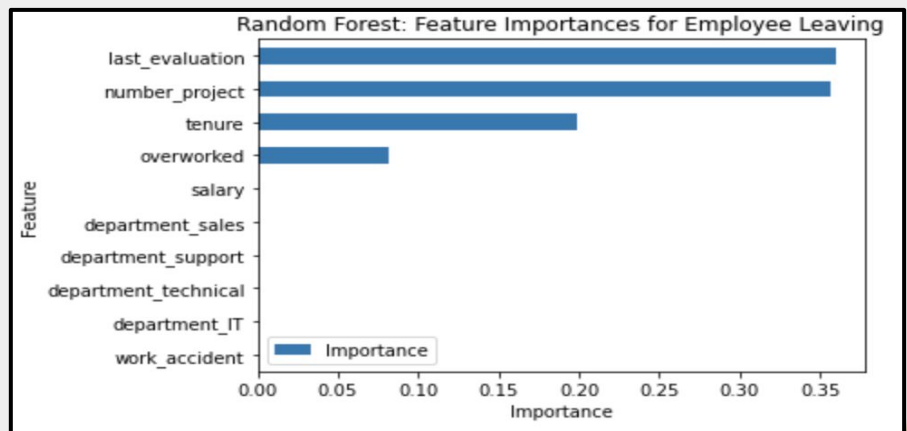
After evaluating the models, it was determined that the Random Forest model was the champion model, outperforming the Decision Tree model.

IMPACT

The model helps predict whether an employee is likely to leave and point to which factors are most influential. Those insights can be leveraged by HR to make decisions that can increase employee retention.



The barplot above shows that in this Decision Tree model, last_evaluation, number_project, tenure, and overworked have the highest importance, in that order. These variables are most helpful in predicting the outcome variable, left.



The plot above shows that in this Random Forest model, last_evaluation, number_project, tenure, and overworked have the highest importance, in that order. These variables are most helpful in predicting the outcome variable, left, and they are the same as the ones used by the Decision Tree model.

KEY INSIGHTS

- Limit the number of projects that employees can work on.
- Consider promoting employees who have been with the company for at least four years, or further investigate the factors contributing to dissatisfaction among employees with this level of tenure.
- Either provide appropriate compensation for extended working hours or avoid making long hours an expectation.
- Ensure employees are clearly informed about overtime pay policies. If expectations regarding workload and time off are unclear, establish and communicate them explicitly.
- Hold company-wide and within-team discussions to understand and address the company work culture, across the board and in specific contexts.
- High evaluation scores should not be reserved for employees who work 200+ hours per month. Consider a proportionate scale for rewarding employees who contribute more/put in more effort.